Quantification: 699_HYP_HP_z20c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	18
interior lacuna count	15
frame-edge lacuna count	3

Interior lacunae only: mean, median and sample SD (n = 15)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	1943.2	1511	1592.19	px^2
major_axis_length_px	lacuna shape	86.2112	77.2677	44.3621	px
minor_axis_length_px	lacuna shape	35.3795	32.0301	17.5187	px
aspect_ratio	lacuna shape	2.7213	2.6732	1.2481	unitless
eccentricity	lacuna shape	0.8655	0.9274	0.1191	unitless
solidity	lacuna shape	0.7587	0.7799	0.1105	unitless
roots_count	headline	6.0667	5	3.575	count
ring_length_r30_px	headline	217.867	192	151.518	px
ring_length_r60_px	reported	469.667	405	340.02	px
owned_length_px	ownership-dependent	522.734	185.539	546.411	px
edge_count	ownership-dependent	27.8	16	28.451	count
mean_edge_length_px	ownership-dependent	18.3802	18.924	5.1931	px
perimeter_px	normalised	265.375	224.528	143.597	px
ring_area_r30_px2	normalised	8241.13	7889	3464.34	px^2
ring_area_r60_px2	normalised	18904.3	20014	7072.73	px^2
in_frame_fraction_r30	normalised	0.9842	1	0.0612	unitless
in_frame_fraction_r60	normalised	0.9777	1	0.0862	unitless
ring_density_r30	normalised	0.027123	0.0283982	0.0172983	px^-1
ring_density_r60	normalised	0.024081	0.026819	0.0153076	px^-1
roots_per_100px_perimeter	normalised	2.6751	2.1993	1.6214	per 100 px
ring_attached_length_r30_px	network v2	173.6	139	139.475	px
ring_attached_length_r60_px	network v2	314	182	272.237	px
ring_length_w_r30_px	network v2	242.6	212.297	172.548	px
ring_length_w_r60_px	network v2	525.49	448.363	384.416	px
sholl_crossings_r10	network v2	4.7333	4	3.7506	count
sholl_crossings_r20	network v2	5.6	4	4.2393	count
sholl_crossings_r30	network v2	6.9333	6	4.5429	count
loop_count	loops, report only	1.2	0	2.3664	count
loop_edge_share	loops, report only	0.0833	0	0.1374	share
ring_width_mean_r30_px	width	6.7667	6.8038	0.8995	px

Per field
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measure	value	units
analysed_area_px2	859818	px^2
total_skeleton_length_px	15974	px
canalicular_length_density_per_px	0.0185784	px^-1
junction_count	315	count
junction_density_per_px2	0.00036636	px^-2
lacuna_count	18	count

lacunae_per_px2	2.093e-05	px^-2
skeleton_component_count	295	count
mean_component_length_px	54.1492	px
median_component_length_px	18	px
field_length_density_w_per_px	0.0207754	px^-1
field_density_without_flagged_per_px	0.0141557	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	1026	px
unattached_component_count	51	count
canalicular_length_density_attached_per_px	0.0185784	px^-1
field_length_density_with_unattached_per_px	0.0197716	px^-1
excluded_area_px2	147318	px^2
analysed_field_area_px2	901258	px^2
loop_count	33	count
loop_edge_share	0.1292	share
edge_count	1014	count

Canalicular width, the new measure

measure	value	units
width_mean_px	6.9837	px
width_median_px	7.2111	px
width_p10_px	4	px
width_p90_px	10	px
width_pixels_used	13688	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	582	37.2403	23.3143	1.5973	0.7798	0.8362	-1.3908	9.6323	718.978	True
2	1079	46.6189	34.6236	1.3464	0.6696	0.7599	-0.6851	205.172	387.171	False
3	3222	86.3749	58.8691	1.4672	0.7318	0.7263	-0.9453	298.969	636.32	False
4	4692	101.01	74.3902	1.3578	0.6765	0.6687	0.5065	385.336	36.169	True
5	1329	56.9074	38.8202	1.4659	0.7312	0.7408	-0.6935	386.281	385.088	False
6	574	63.7	12.3761	5.147	0.9809	0.8658	-1.4148	410.084	638.57	False
7	961	72.5694	18.5814	3.9055	0.9667	0.8076	1.4502	539.664	699.95	False
8	1511	109.373	23.0638	4.7422	0.9775	0.6218	1.4725	559.445	447.167	False
9	599	49.8148	18.0062	2.7665	0.9324	0.7799	1.3874	565.563	692.351	False
10	898	39.9158	30.7419	1.2984	0.6378	0.8635	1.334	631.296	609.345	False
11	2133	94.4671	56.7385	1.665	0.7995	0.538	0.9278	716.599	755.109	False
12	2065	77.2677	36.7006	2.1054	0.88	0.8765	-0.6191	783.319	730.515	False
13	2136	110.111	32.0301	3.4377	0.9568	0.7495	-0.4861	842.146	688.921	False
14	7245	127.692	88.342	1.4454	0.7221	0.8447	-1.2489	821.694	971.904	True
15	7049	221.75	78.7671	2.8153	0.9348	0.5533	-1.5418	863.529	297.474	False
16	2325	112.289	29.7925	3.769	0.9642	0.7992	1.5515	897.959	158.373	False
17	2028	91.0023	34.0428	2.6732	0.9274	0.8145	0.9697	910.357	523.35	False
18	1239	61.0069	27.5385	2.2153	0.8923	0.8844	-1.1926	1001.21	883.603	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px

1	1	135	406	17.4853	1	17.4853	6.782
2	8	241	769	1311.32	70	18.7331	8.4329
3	8	343	755	809.838	50	16.1968	6.2878
4	10	357	697	1881.07	103	18.2628	7.1545
5	9	424	1083	1709.11	89	19.2035	7.0353
6	6	192	446	345.238	17	20.3081	6.6589
7	4	219	405	185.539	16	11.5962	6.8391
8	16	545	972	1116.52	59	18.924	7.842
9	5	141	237	166.255	8	20.7819	7.3126
10	7	348	751	1017.65	49	20.7683	6.0891
11	8	112	254	103.77	7	14.8242	6.8038
12	2	40	40	52.6274	4	13.1569	5.7532
13	5	73	124	46.2132	5	9.2426	6.3638
14	1	26	67	19.4853	1	19.4853	5.3649
15	5	330	688	680.801	31	21.9613	7.6111
16	3	121	254	120.284	6	20.0474	7.602
17	3	104	205	157.368	5	31.4735	5.7226
18	2	35	62	18.4853	1	18.4853	5.146

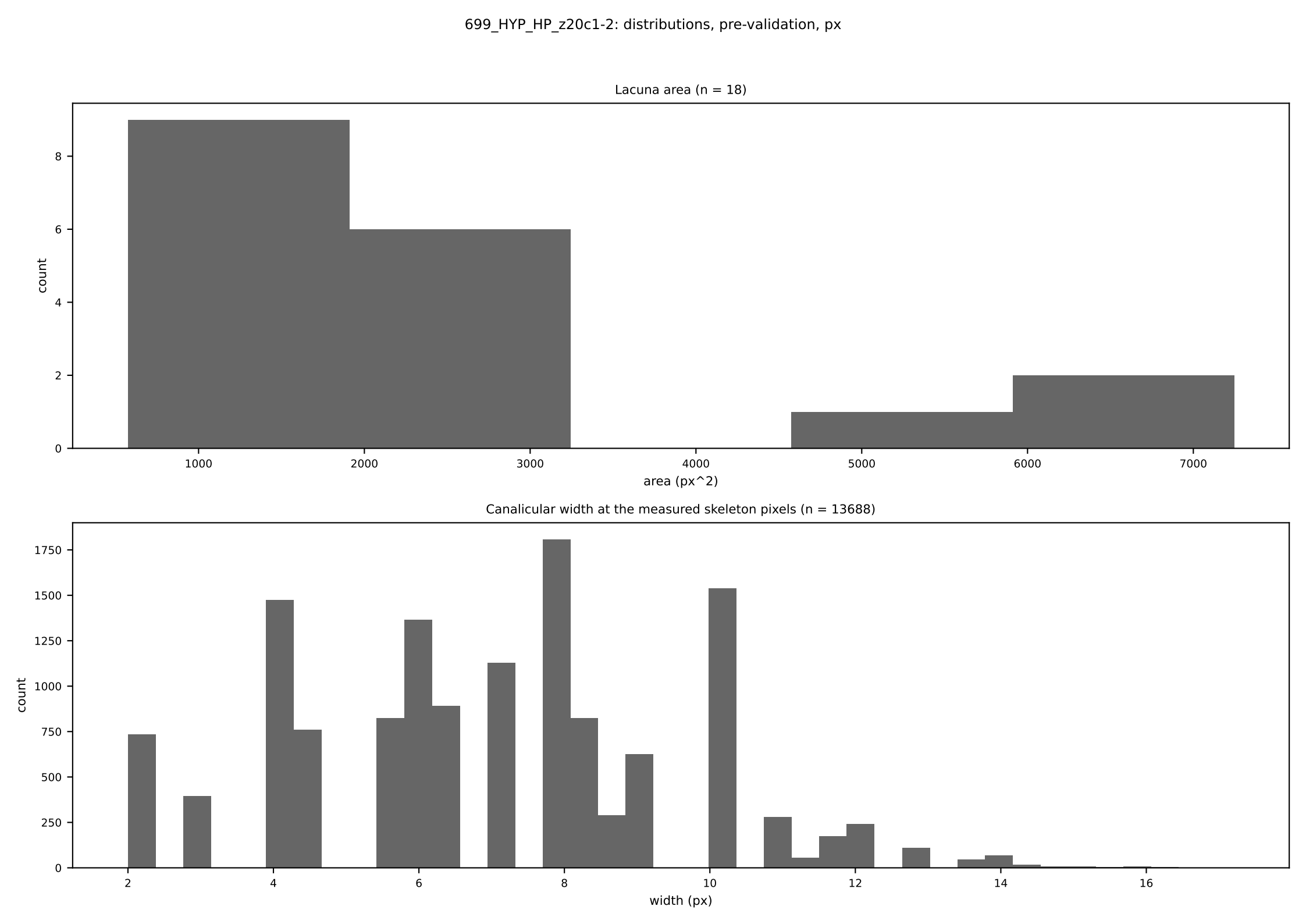
Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	115.083	3546	9848	0.6007	0.566	0.0380711	0.0412266	0.8689
2	185.581	7275	20014	1	1	0.0331272	0.0384231	4.3108
3	418.517	11182	24664	1	1	0.0306743	0.0306114	1.9115
4	534.86	10360	21855	0.7334	0.6731	0.0344595	0.031892	1.8696
5	221.179	7889	21117	1	1	0.0537457	0.0512857	4.0691
6	140.326	6761	16630	1	1	0.0283982	0.026819	4.2758
7	165.74	5442	13737	1	1	0.0402426	0.0294824	2.4134
8	316.149	10460	25779	1	1	0.0521032	0.0377051	5.0609
9	131.439	3919	7916	1	1	0.0359786	0.0299394	3.8041
10	132.267	6479	16157	1	1	0.053712	0.0464814	5.2923
11	363.747	9538	20842	1	1	0.0117425	0.0121869	2.1993
12	224.528	4228	7038	1	1	0.00946074	0.00568343	0.8908
13	285.22	8488	19536	1	1	0.00860038	0.00634726	1.753
14	652.181	10802	23498	0.7216	0.6773	0.00240696	0.00285131	0.1533
15	676.281	17978	35439	1	1	0.0183558	0.0194136	0.7393
16	308.551	9405	21041	1	1	0.0128655	0.0120717	0.9723
17	247.421	8982	20148	1	1	0.0115787	0.0101747	1.2125
18	163.681	5591	13507	0.7628	0.6662	0.00626006	0.00459021	1.2219

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	66	72	153.054	450.291	3	5	5
2	203	665	268.966	859.097	4	6	10
3	216	332	386.433	843.11	9	9	12
4	316	484	394.522	777.843	9	5	8
5	353	769	474.262	1215.65	7	11	11
6	184	253	212.297	498.818	3	6	4
7	102	105	233.054	448.363	5	4	7
8	543	767	624.014	1112.56	15	13	12
9	139	164	151.912	260.338	5	4	4
10	298	637	387.25	841.541	6	12	13
11	112	182	120.498	274.238	4	2	5
12	40	40	42.5563	42.5563	1	1	0

13	73	91	82.7696	147.024	1	2	2
14	21	21	29.1421	74.7696	1	0	2
15	190	450	369.049	768.884	6	9	13
16	79	160	129.083	273.853	3	3	6
17	48	71	116.397	225.48	2	2	4
18	24	24	40.4558	70.8406	0	0	1


699_HYP_HP_z20c1-2: distributions, pre-validation, px
Lacuna area (n = 18)
count
area (px^2)
0
2
4
6
8
1000
2000
3000
4000
5000
6000
7000
Canalicular width at the measured skeleton pixels (n = 13688)
count
width (px)
0
250
500
750
1000
1250
1500
1750
2
4
6
8
10
12
14
16